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CODEX OF THE LOAD

*The official dose of activity, the scale that measures it, the
lifting equation, and what happens to a body with no weight
on it*

VILLUMINATIONS

Fine edition, 2026

Original work · First edition, 2026

villuminations.com

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CHAPTER 1

THERE IS A DOSE, AND IT IS WRITTEN DOWN

There is a physical activity figure set by a government, published in a free document several hundred pages long, reviewed by a scientific committee and last updated in 2018. Almost everyone has heard its headline and almost nobody has read what lies underneath, which is where the useful things are.

WHAT THIS BOOK CLAIMS

Training has a known minimum dose, a particularly profitable stretch and a curve shape. Knowing those three things completely changes the decisions of someone starting out, because most of the benefit is not where people think it is: it is in the first minutes, not the last ones.

THE DOSE, IN A TABLE

MODERATE ACTIVITY	from 150 to 300 minutes a week
VIGOROUS ACTIVITY	from 75 to 150 minutes a week
EQUIVALENCE	one vigorous minute counts as 2 moderate ones
STRENGTH	at least 2 days a week, all the major muscle groups

WHAT THIS BOOK DOES NOT CLAIM

- It is not a personalised training programme. It is the frame inside which any reasonable programme falls, and a way of judging the ones you are offered.
- It promises no aesthetic results and no deadlines. No figure here has a date attached to it, and any plan that does have one is making it up.
- It does not replace a professional. If you have a known cardiovascular, respiratory or metabolic condition, if you are pregnant, if you have had surgery recently or if you have been months without activity, the conversation starts in a consulting room and not in a book.
- It does not deal with high-performance training. The guidelines are written for the health of the population, and that is exactly their value: they are the base on which everything else is built.

THE WARNING SIGN, EXPLAINED JUST ONCE

Chest pain, tightness, pain running down the arm or into the jaw, breathlessness out of proportion to the effort, dizziness or irregular palpitations during exercise: you stop and you seek advice the same day. No session is worth the doubt. It is the one sentence in this book to be remembered by heart.



WHERE EVERYTHING THAT FOLLOWS COMES FROM

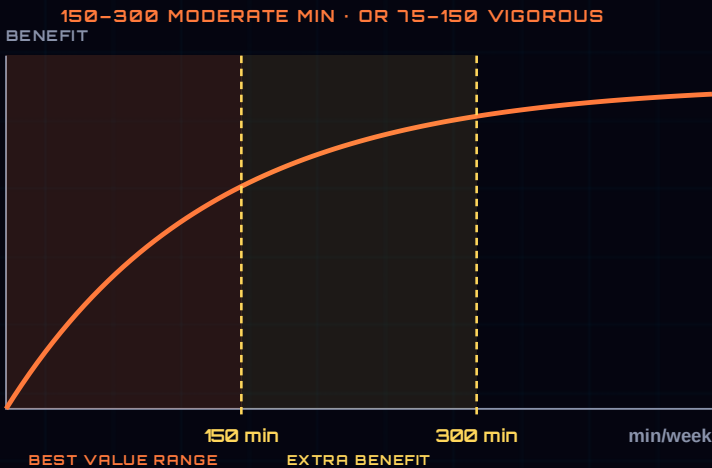
From four public sources of the United States federal government, which by law are not subject to copyright: the physical activity guidelines, the Army's field manual on health and physical readiness, the lifting equation of the national occupational safety institute, and the space agency's research into what happens to a body when the weight is taken off it. All four are named in the verification chapter, with a note on where to download them.

And one more thing: the figures on these pages were not typed into the text. They live in a data module that is verified before the book is typeset. If the equivalence between moderate and vigorous minutes did not add up, or if some activity on the metabolic scale fell outside the intensity band it declares, the book would not be written.

CHAPTER 2

THE SHAPE OF THE CURVE

The guidelines' headline is the figure of a hundred and fifty minutes. The part almost nobody quotes is how the benefit behaves across the range, and that shape is the most actionable information in the whole document.



SHAPE OF THE CURVE PER THE GUIDELINES · NOT A FIT TO DATA

THE RESPONSE CURVE: THE PROFITABLE STRETCH IS AT THE START The benefit does not grow in a straight line with the minutes: it rises very fast at first and then flattens out. The zone marked up to a hundred and fifty minutes is the one with the greatest return per minute invested; beyond three hundred the benefit goes on growing, but each added minute returns less than the one before. The curve draws the shape the guidelines describe, not a fit to particular data.

THE THREE READINGS OF THIS PLATE

FIRST

There is no threshold below which there is no benefit. The curve starts rising from minute one, and the biggest jump in the whole graph happens between doing nothing and doing a little.

SECOND

The most profitable stretch is the one from zero to the recommendation. Someone on forty minutes a week who goes up to ninety gains more than someone on three hundred who goes up to three hundred and fifty.

THIRD

Going beyond the recommendation is not bad. Above the top of the range the benefit goes on rising; it simply stops paying off as much, and the cost in time and in injury starts to weigh.

From this comes the most counter-intuitive piece of advice in the book for someone starting out: do not design the week you would like to do. Design the one you will do even when things go wrong, because the first minutes are the ones worth most and they are precisely the ones lost when the plan is too ambitious and gets abandoned in week three. How much gets done weighs more than what gets chosen, and that is why the chapter on the week gives three templates of different sizes instead of one.



WHAT CHANGED IN THE SECOND EDITION, AND ALMOST NOBODY KNOWS

For years it was recommended that activity be accumulated in bouts of at least ten minutes, and that rule was repeated so often that it is still alive in apps and in gyms. The second edition of the guidelines removed it: all activity now counts, from the first minute.

The change looks administrative and it is not. It makes valid the stairs instead of the lift, the seven minutes walking to the next stop, the four minutes carrying the shopping up. For a person short of time, that repealed rule was what made the recommendation unreachable, and its disappearance is probably the most useful news in the whole document.

- The message about time spent sitting was also strengthened: reducing it has a benefit of its own, and part of the risk associated with sedentary behaviour is attenuated by activity, although it is not removed entirely.
- And it was underlined that some benefits appear immediately—a single session improves that night's sleep, anxiety, insulin sensitivity and blood pressure for hours—and not only after months of consistency. It is a different argument from the usual one and it works better with people who do not believe in long timescales.

CHAPTER 3

THE CURRENCY OF INTENSITY

To be able to add up minutes you have to know which ones count, and for that there is a unit: the metabolic equivalent. One MET is the cost of sitting at rest. Everything else is expressed as a multiple of that.



THE METABOLIC SCALE WITH ITS TWO OFFICIAL THRESHOLDS

One MET is the cost at rest. Activity counts as moderate from 3.0 MET and as vigorous from 6.0. Those two thresholds are the ones the guidelines use to classify, and they are the reason a gentle stroll does not count and a brisk walk does.

THE TWO THRESHOLDS

LIGHT

below 3.0 MET. It counts towards reducing time spent sitting, not towards the weekly dose.

MODERATE

from 3.0 to 6.0 MET. It is the currency of the recommendation.

VIGOROUS

from 6.0 MET upwards. Each minute is worth 2.

THE TALK TEST, WHICH NEEDS NO WATCH

The thresholds have a practical translation that works surprisingly well and needs neither a heart rate monitor nor a table:

- If you can talk but not sing, you are in moderate.
- If you cannot say more than a few words in a row without taking a breath, you are in vigorous.
- If you can sing comfortably, you are in light and it does not count towards the dose.

This test has an advantage over the value in the table: it is relative to whoever is doing it. METs are a population average, and walking at five kilometres an hour is moderate for the average adult and vigorous for someone very untrained or advanced in years. The intensity that counts for health is the relative one, and the guidelines say so explicitly.



WHAT A MET IS REALLY FOR

It serves to estimate the cost, because one MET is roughly one kilocalorie per kilo of body weight per hour. The complete formula fits on one line: MET times kilos times hours.

HALF AN HOUR, IN A PERSON OF 70 KILOS

BRISK WALKING · 5 KM/H	3.5 MET · about 122 kcal · moderate
WEIGHTS, GENERAL SESSION	3.5 MET · about 122 kcal · moderate
CYCLING · 20 KM/H	8 MET · about 280 kcal · vigorous
JOGGING · 8 KM/H	8 MET · about 280 kcal · vigorous
RUNNING · 11 KM/H	11 MET · about 385 kcal · vigorous

WHY THESE FIGURES ARE SMALLER THAN YOU EXPECTED

Because almost every gym machine exaggerates, and because the number that matters is the NET cost —what is spent over and above sitting— and not the gross one. Half an hour of decent running comes to around four hundred kilocalories in a person of seventy kilos, not the eight hundred on the display. This is the arithmetic that explains why trying to offset food with exercise almost always goes badly: the body spends slowly and eats fast.

CHAPTER 4

THE HEART AND THE ZONE THAT DOES NOT EXIST

Any gym machine draws on its screen a band labelled «fat burning zone», always below the band next to it. It is true and useless at the same time, and the mixture of the two makes it the most widespread piece of misinformation in training.

WHAT IS TRUE

At low intensity, the proportion of energy the body draws from fat is greater. At high intensity, that proportion falls and the carbohydrate share rises. That is well established and it is what the screen draws.

WHAT THE SCREEN DOES NOT DRAW

That what matters is not the proportion but the total amount. Half an hour at low intensity burns a high percentage of a small cake; half an hour at high intensity burns a lower percentage of a considerably bigger cake, and the slice is usually larger. Add to that the fact that the raised cost carries on after intense effort, and that the day's accounting does not distinguish where each kilocalorie came from.

The conclusion is not that you must always train hard. It is that the choice of intensity is made for other reasons —time available, recovery,

taste, joints— and not for the substrate. Either of the two bands on the screen works; the one you keep up over time works better.



THE FORMULA EVERYONE USES AND ALMOST NOBODY KNOWS IS BAD

Two hundred and twenty minus your age is the maximum heart rate estimate carried by almost every machine and almost every watch. It is a rule of thumb whose typical error is around ten or twelve beats a minute in either direction, and that deviation is wider than the bands the machine itself draws on top of it.

In practice it means that a forty-year-old whose real maximum is 195 will be training in the wrong zone for the whole session if they follow the number on the screen, which will have calculated 180 for them. That is why the official guidelines do not divide activity into heart rate bands: they classify it by relative intensity, which is what the talk test in the previous chapter measures.

THREE WAYS OF MEASURING INTENSITY, FROM WORST TO BEST FOR THE GENERAL PUBLIC

CALCULATED HEART RATE BANDS

Convenient and systematically inaccurate for anyone whose real maximum does not match the formula, which is almost everyone.

PERCEIVED EXERTION

Scoring from one to ten how hard it feels. Subjective and, precisely for that reason, relative to whoever is training. It correlates quite well with real intensity.

THE TALK TEST

Talking yes, singing no, is moderate; short phrases only, is vigorous. Free, immediate, and it matches the official thresholds.

WHAT THE HEART RATE MONITOR IS FOR, THEN

For two good things. The first, comparing sessions against yourself over the months: if the same route at the same pace comes out at fewer beats, you have improved, and that is a clean piece of data whatever your maximum happens to be. The second, spotting irregularities that deserve a consultation. What it is not for is telling you which band you should be in today.

STRENGTH IS NOT OPTIONAL

In the list of official recommendations there is a line quoted far less often than the one about minutes and which sits at exactly the same level of hierarchy: strength work for all the major muscle groups, at least two days a week.

It is not an add-on for whoever fancies it, nor a second-class recommendation. It appears in the same block as aerobic activity and is not replaced by it: running more does not cover the strength requirement, and lifting weights does not cover the aerobic one. They are two different adaptations of two different systems.

WHAT COUNTS AS STRENGTH WORK

THE MAJOR GROUPS, AND ONE OPTION FOR EACH

LEGS	Squat, leg press, lunge, stair climbing with a load, glute bridge
HIP AND LOWER BACK	Deadlift in any variant, hip hinge, hip extension
CHEST	Bench press, push-ups, dips, dumbbell press
BACK	Row, pull-ups, lat pulldown, band row
SHOULDERS	Overhead press, raises
ARMS	Curl, triceps extension. They get a lot of indirect work: they are the ones that need the least specific attention
ABDOMEN AND TRUNK	Plank, anti-rotation work, loaded carries

The equipment is beside the point. The guidelines draw no distinction between barbell, dumbbell, machine, resistance band or body weight: what counts is that the muscle works against a resistance that represents a real effort. Someone who never sets foot in a gym can meet the whole recommendation on the floor at home, and that is a piece of information the fitness industry has no incentive to spread.



THE ONE VARIABLE YOU HAVE TO RESPECT: PROGRESSION

A muscle adapts to what is demanded of it, and stops adapting when the demand stops rising. Progression does not have to be fast or steady, but it has to exist, and there is more than one way of getting it:

- More weight for the same repetitions.
- More repetitions with the same weight.
- More sets of the same exercise.
- The same work with less rest between sets.
- The same work with better technique and more range, which is the most underrated progression and the one that pays best in the first months.

And a rule of common sense that is in no official document but that prevents most of the injuries of people starting out: raise one variable at a time, and raise it a little. Haste in the first weeks is the commonest cause of stopping in the following ones, and stopping costs far more than going slowly.

WHAT MILITARY DOCTRINE SAYS ABOUT THIS

The United States Army's health and physical readiness manual — published in full and downloadable— organises training in phases and devotes a notable number of pages to gradual progression and to the prevention of overuse injuries. An institution that trains hundreds of thousands of people at once cannot afford to injure them, and that is why its doctrine is considerably more conservative than the usual gym talk. It is worth reading for that.

CHAPTER 6

THE EIGHT PATTERNS

A strength programme is not organised by muscles: it is organised by movement patterns. There are eight, they cover everything a human body does against a resistance, and a week that touches them all leaves no gaps by definition. It is the quickest way to audit any plan, including one being sold to you.

1	SQUAT Knee and hip bend together	Squat, leg press, lunge, step-up...
2	HIP HINGE The hip leads and the knee barely bends	Deadlift, good morning, hip thrust...
3	HORIZONTAL PUSH Push weight from the chest	Bench press, push-ups, dips
4	VERTICAL PUSH Take weight above the head	Overhead press, dumbbell press
5	HORIZONTAL PULL Draw weight to the torso	Barbell row, dumbbell row...
6	VERTICAL PULL Draw body or weight from above	Pull-ups, lat pulldown
7	CARRY Hold a load and walk	Farmer's walk, suitcase carry...
8	ANTI-ROTATION Resist a twist without twisting	Plank, Pallof press, suitcase...

EIGHT PATTERNS · A WEEK THAT TOUCHES THEM ALL LEAVES NO GAPS

THE EIGHT PATTERNS AND ONE OPTION FOR EACH Organised by what the joint does and not by the muscle you want to «work». Any exercise you know of falls into one of the eight, and if your week leaves one of them empty, there is your gap.

THE EIGHT, WITH EXAMPLES

SQUAT	Knee and hip bend together · Squat, leg press, lunge, step-up
HIP HINGE	The hip leads and the knee barely bends · Deadlift, good morning, hip thrust, swing
HORIZONTAL PUSH	Pushing weight away from the chest · Bench press, push-ups, dips
VERTICAL PUSH	Taking weight above the head · Overhead press, dumbbell press
HORIZONTAL PULL	Drawing weight towards the torso · Barbell row, dumbbell row, inverted row
VERTICAL PULL	Drawing the body or the weight from above · Pull-ups, lat pulldown
CARRY	Holding a load and walking · Farmer's walk, suitcase carry, front carry
ANTI-ROTATION	Resisting a twist without twisting · Plank, Pallof press, one-sided carry

WHY THIS WAY OF ORGANISING AND NOT THE MAGAZINES'

The classic split by muscle groups —Monday chest, Tuesday back — comes from competitive bodybuilding and serves its purpose reasonably well, which is to hypertrophy particular areas with a lot of volume and a lot of weekly gym time. For almost everyone, that is not the purpose.

Organising by patterns has three practical advantages. The first: gaps are spotted at a glance, and the typical gap of a whole gym is the hip hinge and the carry. The second: each pattern allows dozens of

variants, so an injury or an occupied machine does not break the session. And the third: they are the movements life asks for outside the gym, which is literally what all of this is for.



THE COMMONEST GAP, AND THE MOST EXPENSIVE

The hip hinge. It is the pattern of picking something up off the floor, and it is the one most people avoid because the deadlift has an undeserved bad name. The consequence is predictable: a weak posterior chain in a population that spends the day sitting, in the very movement in which most back scares occur. The NIOSH equation in the next chapter explains why training it matters so much.

The second gap is the carry: holding weight and walking. It appears in almost no programme, it requires no technique and it trains grip, trunk and posture at once. The heavy supermarket shop is the functional exercise nobody had to invent.

HOW TO AUDIT A WEEK IN TWO MINUTES

Write the eight patterns in a column and put the exercises you did last week next to them. The ones left empty are your real programme, and they almost never match the one you think you are following.

CHAPTER 7

THE SIX VARIABLES

A training session has only six things that can be moved. Knowing them serves two purposes: progressing when the weight will not go up, and not making the commonest mistake of all, which is raising two at once.

SIX VARIABLES · YOU RAISE ONE AT A TIME

1 INTENSITY

How much it weighs against your max, or how many reps you had left in reserve when the set ended

2 VOLUME

Effective sets per muscle and per week. The count that actually predicts the gain in size

3 FREQUENCY

How many times a week each muscle receives a stimulus

4 DENSITY

How long you rest between sets. It changes the stimulus without changing the weight

5 RANGE OF MOTION

How much joint range you use. The cheapest variable and the one most often sacrificed to vanity

6 TEMPO

How fast you lower and lift. The lowering phase is where most adaptation is created

RAISING TWO AT ONCE IS HOW PEOPLE GET HURT

THE SIX TRAINING VARIABLES Each one alters the stimulus on its own. You raise one and hold it for a few weeks; raising two at once multiplies the real load without your noticing, and it is the injury factory of month two.

THE SIX

INTENSITY	How much it weighs against your maximum, or how many repetitions you had left in reserve when the set ended
VOLUME	Effective sets per muscle per week. The count that really predicts the gain in size
FREQUENCY	How many times a week each muscle receives a stimulus
DENSITY	How long you rest between sets. It changes the stimulus without changing the weight
RANGE OF MOTION	How much joint range you use. The cheapest variable and the one most often sacrificed to vanity
TEMPO	How fast you lower and lift. The lowering phase is where most adaptation is created

THE ONE THAT PAYS BEST IN THE FIRST MONTHS

Range of motion. A half squat with twice the weight is not more training than a full squat with half of it: it is a different exercise, and a considerably worse one. Full range produces more adaptation in the part of the muscle that works stretched, maintains the mobility of the joint and, along the way, forces you to use an honest weight.

It is also the variable most often sacrificed to vanity, because dropping the weight in order to complete the range feels like going backwards. It is not. It is the only way for the number you write down to mean anything.

THE ONE ALMOST NOBODY COUNTS PROPERLY: VOLUME

The count that predicts the gain in size is effective sets per muscle per week, not the time spent in the gym or the number of exercises. An effective set is one taken close to technical failure: with four or five repetitions of margin left over, the stimulus is thin however many sets are done.

Hence a tightly packed forty-five-minute session returns more than two hours of drifting between machines, and hence too that raising volume without limit stops working: a point arrives at which what is added is not recovered, and what is not recovered does not adapt.



THE EFFORT SCALE, WHICH SOLVES THE PERCENTAGE PROBLEM

Programming by percentage of the maximum requires knowing the maximum, and the maximum changes with sleep, with stress and with the week. The practical alternative is to score how many repetitions you had left in reserve when the set ended:

REPETITIONS IN RESERVE

0	You could not have done one more. Used sparingly and with judgement
1 A 2	The working zone for strength and size sets
3 A 4	A valid stimulus at much less recovery cost
5 OR MORE	Warm-up, technique or filler volume

With this scale a bad day adjusts itself: the weight that leaves you two repetitions short today may be lower than last week's, and the session was still correct. Programming by a fixed percentage forces you either to fail or to lie.

THE PROGRESSION RULE THAT PREVENTS ALMOST EVERYTHING

One variable, one step, and hold it for two or three weeks before touching another. If performance drops for three sessions running, or if joint pain appears, the previous step was the right one: you go back and hold it longer. Going down is not going backwards; it is what makes it possible to keep going up next year.

CHAPTER 8

WARMING UP, COOLING DOWN AND WHAT IS NO USE

The warm-up is the part of training with the most mythology per minute invested. What follows separates what has reasonable backing from what gets repeated out of habit.

A WARM-UP THAT ACTUALLY DOES SOMETHING, IN THREE ACTS

- 1** Five or ten minutes of general movement that raises temperature and heart rate. Brisk walking, easy cycling, light skipping. Anything that is not stretching.
- 2** Active mobility of the joints you are going to use. Circles, swings, bodyweight squats, rotations. Movement, not held positions.
- 3** Ramp-up sets of the first exercise: the empty bar, then half the weight, then something just below the working weight. It is the part of the warm-up that returns most and the part most people skip.

The aim is not to prevent injuries —the evidence on that is weaker than is claimed— but to perform better in the first sets, which is an immediate and checkable effect: the first set after warming up properly feels different, and that alone justifies the ten minutes.



STATIC STRETCHING, PUT IN ITS PLACE

- Before training, holding a long stretch slightly reduces the strength available over the following minutes. If you do it, keep it brief and not immediately before the heavy set.
- It does not demonstrably prevent injuries, and we have spent decades trying to demonstrate that it does.
- It neither prevents nor cures muscle soreness. This is quite well established and it still turns up everywhere.
- It does improve range of motion if done regularly, and that has a value of its own. Do it because you want more mobility, not because you believe it protects you.

MUSCLE SORENESS, WHICH IS NOT WHAT PEOPLE THINK

They appear above all after eccentric work —the lowering phase— and after new exercises, they peak between twenty-four and forty-eight hours later and they go away on their own. They are not an indicator that the session was good: a well-run programme produces less and less soreness while performance keeps rising, and chasing them is chasing the wrong signal.

What relieves them best is gentle movement. What does not shift them is stretching. And if they last more than four or five days, or if they come with marked swelling and dark urine after an unusually hard session, that is no longer soreness and it is a reason to seek advice the same day.

COOL DOWN

Easing off for five minutes at the end does not appreciably speed up recovery, but it avoids the dizziness of stopping dead after intense effort and it is a good moment for mobility work, because the tissue is warm. It is a cheap habit with two small advantages; do not ask more of it.

AEROBIC WORK FROM THE INSIDE

«Doing cardio» describes as little as «eating food». Under that label fit three different things that produce different adaptations and cost different recoveries, and knowing which is which changes the outcome of a season.

THE THREE KINDS

BASE	Conversation possible throughout the session. Comfortable volume, from thirty minutes to two hours. It builds background capacity and costs almost no recovery.
THRESHOLD	A sustained pace at which sentences get shorter. Blocks of ten to twenty minutes. It improves the pace you can hold for a long time.
INTERVALS	Short, very intense efforts with rest. They raise the ceiling, cost a good deal of recovery and are the most overused of the three.

THE MISTAKE OF LIVING IN THE MIDDLE

The commonest trap of amateur aerobic work is doing it all at an intermediate intensity: too hard to accumulate volume without fatigue and too easy to produce the adaptation of high intensity. It is comfortable, it feels productive and it returns little for what it costs.

The correction is simple and it works: let most of the volume be clearly easy —genuinely easy, talking comfortably— and let the hard part be clearly hard and take up little. The extremes do the work; the middle accumulates fatigue.



DOES AEROBIC WORK INTERFERE WITH STRENGTH?

The question has been in the air for decades and the reasonable answer is: far less than the gym fears, and rather more than zero. What is known with some solidity is that the interference appears above all with very high aerobic volumes, with both things in the same session and with little recovery between them.

- If you can, separate the two sessions by several hours or put them on different days.
- If they have to go together, do first whatever is the priority for you.
- Running interferes more than cycling, probably because of the eccentric component of the stride.
- For the vast majority of people, who do three or four hours a week in total, this discussion is purely theoretical.

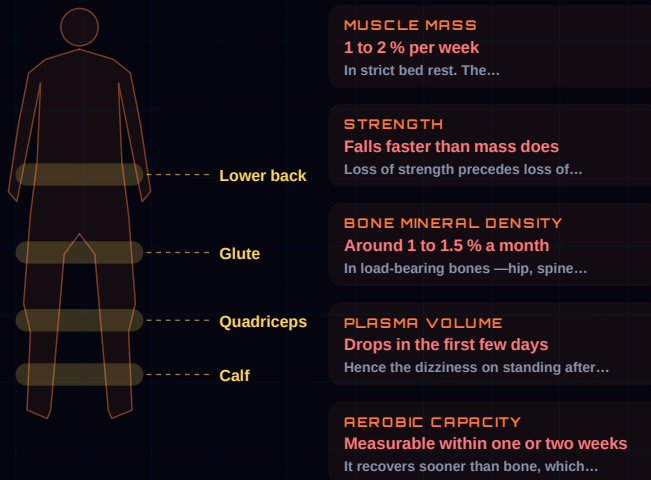
WHAT AEROBIC WORK DOES AND STRENGTH DOES NOT

It changes the heart and the vascular system in a way strength training does not reproduce, and those changes are the ones most strongly associated with living more years and living them better. That is why the guidelines ask for both and do not let you choose: they are not substitutes.

CHAPTER 10

THE EXPERIMENT OF TAKING THE WEIGHT AWAY

To find out what load is for, the cleanest thing is to remove it entirely and watch what happens. That is exactly what spaceflight and prolonged bed rest research programmes have been doing for decades, and their results are publicly accessible.



BED REST AND SPACEFLIGHT · ORDERS OF MAGNITUDE

WHAT HAPPENS TO A BODY WITH NO LOAD ON IT The areas marked are the ones that lose most: what holds the body up against gravity. The column gives the rate of each loss.

These are orders of magnitude from the bed rest and spaceflight literature. The studies differ on the figure according to duration, population and method of measurement; what they do not differ on is the sign, and that agreement is what makes the whole reliable.

THE FIVE EFFECTS

MUSCLE MASS	1 to 2 % a week · In strict bed rest. The leg and back extensors are the first to go and the ones that lose most
STRENGTH	It falls faster than mass · Loss of strength precedes loss of size: part of it is neural
BONE MINERAL DENSITY	Around 1 to 1.5 % a month · In load-bearing bones —hip, lumbar spine, calcaneus—. The arms barely change
PLASMA VOLUME	It drops in the first days · Hence the dizziness on standing up after days in bed
AEROBIC CAPACITY	Measurable in one or two weeks · It recovers sooner than bone, which is the slowest of all

THE ORDER MATTERS MORE THAN THE FIGURES

The first thing to go is not muscle: it is strength. And it goes first because part of strength is neural —the ability to recruit and coordinate fibres— and the nervous system unlearns before the tissue shrinks. That same asymmetry explains why a beginner's first weeks of training produce spectacular strength gains with barely any change in appearance: they are learning, not growing yet.

The last thing to go, and the last to come back, is bone. Mineral density responds to mechanical load with a slowness that has no

comparison with the rest: it is lost over months and regained over more months, and in some cases not entirely. It is the reason strength training is a long-term investment with a component that cannot be improvised at sixty.



AND FROM THIS COMES A GENERAL PRINCIPLE

The tissue that bears the load is exactly the tissue the load maintains. The arms barely lose bone density in bed rest because they did not have it from bearing weight in the first place; the hip, the lumbar spine and the heel lose a great deal because their structure is the result of years of standing up.

What this means for someone who is not going into space is direct: sitting eleven hours a day is a slow, partial version of the same experiment. Not at the same intensity, not at the same rate, but in the same direction. Unloading is not an exotic astronaut phenomenon: it is the technical name for what happens to a sedentary life.

THE PRACTICAL CONSEQUENCE THAT IS IN THE GUIDELINES

Breaking up time spent sitting has a benefit of its own, on top of structured activity. Getting up every half hour, even for two minutes, and replacing part of the sitting time with standing or walking is an explicit recommendation of the second edition. It does not replace the hundred and fifty minutes; it is added to them.

LOSING AND REGAINING

The other face of unloading is what happens when you stop training for a few weeks, which is a far commoner situation than a spaceflight: flu, a house move, an exam, a long trip, an injury to something else.

THE REAL SCALE OF DETRAINING

WHAT IS LOST AND HOW LONG IT TAKES

FIRST DAYS	Plasma volume falls and with it part of the apparent aerobic capacity. It is the fastest change and also the fastest to come back.
ONE OR TWO WEEKS	A measurable drop in aerobic capacity. Strength, on the other hand, holds up quite well over this period.
THREE TO FOUR WEEKS	The loss of strength starts to show. Muscle size holds out longer than people fear.
MONTHS	Here there is appreciable loss of mass, and bone density starts to move, which is the slowest of all.

The practical conclusion is reassuring and worth saying, because fear of losing everything is why a lot of people train injured: two weeks off destroys nothing. Neither does a month. What destroys progress is

training on top of an injury until it becomes chronic, and then the months really do disappear.



THE RESTART, WHICH IS WHAT ALMOST EVERYONE GETS WRONG

On coming back the sensation is deceptive: strength returns quickly—part of it was neural and gets relearned—while tendons, fascia and connective tissue adapt far more slowly than muscle. That lag is a tendinopathy factory. The muscle can already pull a weight the tendon is not yet ready to bear.

- 1 Come back with half the volume you were doing, not with the same weight.
- 2 Raise one step a week for three or four weeks until you reach the starting point.
- 3 If the break was due to injury or illness, the step is set by whoever treated you, not by this book.
- 4 Expect disproportionate soreness in the first week and do not read it as a sign of anything, unless it lasts more than three or four days or comes with dark urine, which is a reason to seek advice immediately.

THE PHENOMENON THAT WORKS IN YOUR FAVOUR

Regaining is faster than building for the first time. Someone who once had muscle mass regains it sooner than someone who never had it, and this is observed consistently. It is a good argument against the idea that dropping out for two months throws away two years: it does not.

CHAPTER 12

PAIN: HOW TO TELL IT APART

This chapter exists because the alternative to having it is people learning to tell the difference the expensive way. It does not replace a diagnosis; it gives criteria for knowing when to carry on, when to change and when to stop and seek advice.

THREE THINGS THAT GET CONFUSED

DIFFUSE MUSCULAR DISCOMFORT

Spread through the belly of the muscle, symmetrical, appearing hours later, improving with movement and gone in two or three days. It is normal.

LOCALISED JOINT PAIN

You can point to it with one finger, it always appears in the same movement and at the same point of the range, it does not improve with warming up. It is not normal.

PAIN THAT GROWS SESSION BY SESSION

It starts at the end of the session, then at the beginning, then in daily life. It is the classic script of an overuse injury, and the sooner it is cut short, the less it costs.

WHAT TO DO WITH EACH ONE

- 1** Muscular discomfort: carry on, move gently and do not measure the quality of a session by how much it hurts afterwards.
- 2** Pain in a particular movement: change the movement, do not abandon the pattern. If the bench press hurts, try dumbbells or a

different angle; there is almost always a variant that does not hurt, and carrying on training the pattern is better than dropping it.

- 3 Pain that grows: reduce the load of that pattern by one large step for two weeks, keep the rest and watch. If it has not improved in two weeks, seek advice.
- 4 Any pain with swelling, sudden loss of strength, a locked joint, tingling or loss of sensation: seek advice straight away and with no trial phase.



AND THE SIGNS THAT STOP A SESSION ON THE SPOT

- Pain or tightness in the chest, or pain radiating to the arm, the neck or the jaw.
- Breathlessness out of proportion to the effort you are making.
- Dizziness, blurred vision, irregular palpitations or a feeling of fainting.
- A sudden, intense headache during an effort.
- Dark-coloured urine after an unusually hard session, especially if it comes with disproportionate muscle pain and swelling.

These five are not «listening to your body»: they are grounds for medical attention the same day. No session of any plan in any book is worth the doubt, and this is the page worth remembering out of the whole volume.

THE SEVENTY-TWO-HOUR RULE

If something hurts the same or worse three days after you reduced the load, it has stopped being a matter of programming. What comes next is not choosing another exercise: it is having someone examine you.

THE LIFTING EQUATION

There is an official equation that calculates how much weight it is reasonable to lift in a particular situation. It was published by the United States national institute for occupational safety and health, it is used in workplace inspection across half the world, it is freely available, and practically nobody who trains has ever seen it.

RECOMMENDED WEIGHT = 23 kg × SIX FACTORS

each factor is between 0 and 1 · never improves it



HORIZONTAL

Distance of the load from the body. It...



VERTICAL

Height of the hands at the start



DISTANCE

How far the load travels



ASYMMETRY

How much the trunk twists



FREQUENCY

Lifts per minute and duration of...



COUPLING

Good, fair or poor

MOVING THE LOAD AWAY FROM THE BODY PENALISES MOST

THE RECOMMENDED WEIGHT AND THE SIX FACTORS THAT REDUCE IT

You start from a constant load of 23 kilograms under ideal conditions and multiply by six factors valued between zero and one. No factor can improve the result: only reduce it. The one that penalises most, by a wide margin, is the horizontal distance of the load from the body.

The recommended weight is obtained by multiplying the 23 kg by six factors valued between 0 and 1. Each poor condition reduces the result, and that is why the same weight is safe held against the body and is not safe at arm's length.

THE SIX FACTORS

HM · HORIZONTAL	Distance of the load from the body. It is the factor that penalises most: moving the load away multiplies it
VM · VERTICAL	Height of the hands at the start
DM · DISTANCE	How far the load travels
AM · ASYMMETRY	How much the trunk twists
FM · FREQUENCY	Lifts per minute and duration of the task
CM · COUPLING	Good, fair or poor

THE CONSEQUENCE THAT CHANGES HOW YOU LIFT

Because the factors multiply, two mediocre conditions at once do not add: they compound. A factor of 0.7 and another of 0.7 give 0.49, that is, less than half the recommended weight. That is why lifting something away from the body while twisting the trunk is so disproportionately worse than doing either of the two separately, and why back injuries almost always happen in combined movements and not in big, clean lifts.

Horizontal distance is the main lever. A weight held against the body and one held at arm's length are, mechanically, different loads on

the spine even though the scales say the same thing. That is the entire technical explanation of the folk advice to «hold the box against your chest», and knowing it turns it into something you remember.



WHAT THE EQUATION DOES NOT SAY, AND NEEDS SAYING

It does not say that bending the back is the cause of injuries. The equation contains no lumbar curvature factor; it contains distance, height, travel, twist, frequency and coupling. The «straight back» mantra is a useful simplification for beginners, but the evidence on specific posture as a cause of injury is far less conclusive than the repetition suggests.

What is established is the other thing: the further away, the higher, the more twisted, the more repeated and the worse the grip, the worse. Those five things can be changed in any real situation, and changing them returns considerably more than obsessing over the shape of the spine.

HOW THIS IS USED IN THE GYM

The deadlift and the row carry the load as close to the body as anatomy allows, and that is no aesthetic accident: it is the horizontal factor taken to its optimum. When the bar drifts away from the legs in a deadlift, the exercise becomes, literally, heavier. The equation explains why technique changes what the bar costs without changing what the bar weighs.

HOW A WEEK IS ORGANISED

With the dose, the intensity scale and the strength rule, there is everything needed to build a week. This chapter gives three, according to the time available, and none of them is anyone's in particular: they are templates to be modified.

A THREE-HOUR WEEK · THE MINIMUM THAT MEETS THE MARK

MONDAY	Strength, whole body, 45 min. Legs, push, pull, trunk
TUESDAY	Brisk walking, 30 min. Talk test: talking yes, singing no
WEDNESDAY	Rest or a gentle walk
THURSDAY	Strength, whole body, 45 min. The same patterns, different variants
FRIDAY	Brisk walking, 30 min
SATURDAY	Something you enjoy, 45 min. Cycling, hiking, padel, dancing
SUNDAY	Rest

Total: about 105 minutes of moderate aerobic work plus two strength sessions. It falls a little short of the moderate range and

Saturday's session makes up for it. It is the minimum skeleton that meets both recommendations at once, and it fits into three hours.

A FIVE-HOUR WEEK · THE COMFORTABLE RANGE

MONDAY	Strength, lower body and trunk, 50 min
TUESDAY	Continuous moderate aerobic work, 40 min
WEDNESDAY	Strength, upper body, 45 min
THURSDAY	Aerobic work, 30 min, with a vigorous stretch or two
FRIDAY	Strength, whole body, 45 min
SATURDAY	A long, enjoyable activity, 60 to 90 min
SUNDAY	Rest or mobility

Here you enter squarely into the extra benefit stretch of the curve, with three strength sessions instead of two. It is probably the best balance between results and life for most adults.

AN HOUR-AND-A-HALF WEEK · WHEN THERE IS NO MORE

TWO DAYS	Whole-body strength, 30 min, with no long rests
THREE DAYS	Ten minutes of stairs, a hill or brisk walking
EVERY DAY	Getting up from your seat every half hour. It does not count towards the dose and it counts towards something else

This week does not reach the recommendation, and it is included on purpose. The curve in chapter two says the jump from zero to this is the most profitable in the whole graph; the jump from this to the full recommendation returns less. An honest book has to say that the small version is worth a great deal, because the real alternative is usually not the big version: it is nothing.

ON WARMING UP AND STRETCHING

Warming up works, and it works above all for performing better in the first sets: five or ten minutes of general movement and a few light sets of the exercise you are about to do. Stretching beforehand, on the other hand, does not prevent injuries —the evidence on that is quite poor— and prolonged static stretching immediately before slightly reduces the strength available. If you like stretching, do it afterwards or at another time of day.

TRAINING FROM SIXTY ONWARDS

The physical activity guidelines have a specific section for older adults, and it contains the only recommendation in the whole document that does not appear for the rest of the population. It deserves a chapter because almost nobody knows about it and because it is the one that buys the most years of independence.

WHAT THE GUIDELINES ADD FOR OLDER ADULTS

BALANCE	Balance work as a component in its own right, on top of aerobic work and strength. It is the additional recommendation and it is aimed at what really takes independence away, which is falls.
STRENGTH	Just as necessary as at any age, and more urgent: the loss of muscle mass over the years is what turns a fall into a fracture and a fracture into a different life.
AEROBIC WORK	The same dose, with the rider that anyone unable to reach it should be as active as their condition allows. The bar adapts; it is not cancelled.
ADAPTATION	Adjusting the effort to the level of fitness, with particular attention to chronic conditions. Adapting is not lowering to zero.

BALANCE CAN BE TRAINED, AND IT IS CHEAP

- Standing on one leg while you brush your teeth. Two minutes a day that take up no new time.
- Walking in a line, heel to toe, a few metres down the corridor.
- Getting up from a chair without using your arms, several times in a row. It is strength and balance at once, and it is the test that best predicts independence ten years out.
- Changes of direction and of surface while walking: grass, a slope, sand. Balance is trained by forcing it to correct.
- All of this is done with a support nearby at first. The progression is removing the support, not starting without it.



THE LOSS OF MUSCLE WITH AGE, AT ITS REAL SIZE

From middle age onwards, muscle mass is lost continuously if nothing is done about it, and strength is lost faster still than mass. That process is not a biological fate to be accepted: it responds to strength training at any age, and there are documented programmes producing gains in people over eighty.

The commonest mistake is the opposite of the assumed one: it is not training too hard, it is training too gently. A weight that can be lifted thirty times without effort produces no adaptation at any age.

Progression has to exist here too, with more care over technique and more margin in the rest, but it has to exist.

PROTEIN AT THIS STAGE

The population floor of 0.8 grams per kilo falls short for holding back the loss of mass beyond a certain age, and the geriatric literature works with clearly higher figures. That figure does not come from the reference tables and I say so: it comes from a different literature. It is a conversation worth having with a professional, particularly where kidney function is compromised, in which case the answer may be the opposite.

CHAPTER 16

A BATTERY TO MEASURE YOURSELF WITH, AND IT IS PUBLIC

Body weight swings by several kilos within days through water, salt, glycogen and gut contents, so measuring progress with the scales is reacting to noise. Measuring it with physical tests is far better, and there is a complete battery, described in detail and freely available, which the United States Army publishes in full.

SIX TESTS · SIX DIFFERENT QUALITIES

MAXIMUM DEADLIFT

Three repetitions with the hex bar

Maximum strength of the chain...

MEDICINE BALL THROW

A 4.5 kg ball thrown backwards over the...

Explosive power of the whole...

HAND-RELEASE PUSH-UP

Chest to the floor and hands off on every rep

Endurance of the upper...

SPRINT, DRAG AND CARRY

250 m: sprint, drag a sled...

Anaerobic power and grip

PLANK

Held, with the body in line

Trunk endurance under...

TWO-MILE RUN

3.2 km timed

Aerobic capacity

MINIMUM 60 OF 100 IN EACH · THEY DO NOT OFFSET

PUBLIC PHYSICAL FITNESS TEST BATTERY

SIX TESTS, SIX DIFFERENT QUALITIES The battery is built so that no test can be offset by another: there is a minimum for each one. That rule is the transferable lesson, even if you never do any of the six.

WHAT EACH ONE MEASURES

MAXIMUM DEADLIFT	Three repetitions with the hex bar · it measures maximum strength of the posterior chain
MEDICINE BALL THROW	A 4.5 kg ball thrown backwards over the head · it measures explosive power of the whole body
HAND-RELEASE PUSH-UP	Chest to the floor and hands off on every repetition · it measures upper-body endurance, with no half range
SPRINT, DRAG AND CARRY	250 m: sprint, sled drag, lateral shuffle and a two-weight carry · it measures anaerobic power and grip
PLANK	Held, with the body in line · it measures trunk endurance in isometry
TWO-MILE RUN	3.2 km timed · it measures aerobic capacity

Each test scores from 0 to 100 and you have to get at least 60 in every one; a bad test is not offset by an excellent one. That rule is the transferable lesson.

WHY THIS IS INTERESTING FOR SOMEONE WHO IS NOT IN THE MILITARY

For the design, not for the particular tests. A battery that demands a minimum in six different qualities is saying something the average gym ignores: that a well-prepared body is not the one that stands out in one thing, but the one that fails in none. Someone who lifts a lot and cannot run two miles has a gap, and that gap is not filled by lifting more.

The second reason is that it is a process goal disguised as an outcome. Scoring better in six tests is a clear, measurable target that depends entirely on what you do, unlike body weight, which depends on half a dozen things you do not control.



A HOME VERSION OF THE SAME IDEA

You need no sled and no hex bar. What you need is to choose five or six tests covering different qualities, measure them today and repeat them every three months under the same conditions:

HOME BATTERY

LEG STRENGTH	Squats in one minute, or the weight of your five-repetition set
PUSHING STRENGTH	Maximum push-ups with full technique
PULLING STRENGTH	Pull-ups, or maximum inverted rows
TRUNK	Plank held, in seconds
AEROBIC WORK	Distance in twelve minutes, or time over a fixed route you can repeat
BALANCE	Seconds on one leg with your eyes closed

Write the six numbers down and keep them. In three months you will have something no set of scales gives: six pieces of evidence that the work did something, or six signals of where it did not. And unlike

weight, these numbers do not swing three kilos depending on what you had for dinner.

HOW NOT TO TAKE THIS MEASUREMENT

It is not done on a bad day as a punishment, nor on the same day as a hard session, nor by changing the conditions between one time and the next. It is done rested, with the same protocol, and it is written down even when it comes out worse than last time. A worse result is a fact about the plan, not a verdict on the person.

CHAPTER 17

HOW TO CHECK IT YOURSELF

This book's four sources are free and online. This chapter says which they are and what to look for in each, because a book that cannot be verified cannot be corrected either.

THE SOURCES OF THIS BOOK

DIETARY REFERENCE INTAKES	National Academies · distributed by the NIH Office of Dietary Supplements. RDA, AI and UL for vitamins and minerals.
PHYSICAL ACTIVITY GUIDELINES FOR AMERICANS, 2ND ED.	U.S. Department of Health and Human Services, 2018. Activity dose and the response curve.
DIETARY GUIDELINES FOR AMERICANS	USDA and HHS. Dietary patterns and limits for added sugars, saturated fat and sodium.
FOODDATA CENTRAL	USDA Agricultural Research Service. Food composition, searchable online at no cost.
FM 7-22 HOLISTIC HEALTH AND FITNESS	U.S. Army. Doctrine on physical training, nutrition, sleep and mental readiness, published in full.
THE NIOSH LIFTING EQUATION	National Institute for Occupational Safety and Health, CDC. Calculation of the recommended weight limit in manual handling.
RESEARCH ON UNLOADING AND BED REST	NASA Human Research Program. Effects of unloading on muscle, bone and the cardiovascular system.
REGISTER OF HEALTH CLAIMS	European Commission, with the scientific assessments of EFSA. What may be claimed about a nutrient, and what was applied for and refused.



THE FOUR THAT SPECIFICALLY HOLD THIS BOOK UP

CONCRETE ROUTES

THE DOSE OF ACTIVITY	Physical Activity Guidelines for Americans, second edition, at health.gov . Look for the adults chapter and the summary of the advisory committee's scientific report, which is where the discussion of the response curve is.
THE TRAINING DOCTRINE	FM 7-22, Holistic Health and Fitness, in the United States Army publications library. Published in full. Its five domains of readiness—physical, nutritional, mental, sleep and spiritual— explain why an impeccable training plan does not deliver when any of the other four fails.
THE LIFTING EQUATION	NIOSH, at cdc.gov/niosh . Look for the revised lifting equation and its applications manual, which comes with worked examples.
UNLOADING	NASA's Human Research Program. Look for the bed rest studies and the documented risks of muscle and bone loss; they are described in the open.

AND HOW THIS BOOK CHECKS ITSELF

The figures on these pages live in a data module that the typesetting program verifies before writing. For this book, the relevant checks are two:

- That the equivalence between intensities adds up at both ends: $75 \times 2 = 150$ and $150 \times 2 = 300$. If either of the two failed, the book would

not be typeset.

- That every activity on the metabolic scale falls inside the intensity band it declares, against the official thresholds. This check caught a real error during the writing: I had classified an intensity as moderate when its value put it right on the threshold of vigorous. The datum was corrected, not the check.

What is more, the energy costs appearing in the metabolic scale chapter are not written by hand: they are calculated from the same table that draws the plate. There is no way for the text and the drawing to say different things.

THE LOAD NOTEBOOK

Four weeks. It is not a training plan: it is a measurement. It serves to find out where you are on the curve in chapter two, which is a fact almost nobody has and which decides what to do next.

WEEK 1 · COUNTING WHAT YOU ALREADY DO

- 1 Write down each day the minutes of activity that pass the talk test. Only the ones that pass it.
- 2 Write the strength minutes down separately, of whatever kind.
- 3 Write down a rough estimate of hours spent sitting. Precision is not needed: it is enough to know whether it is six or twelve.
- 4 Change nothing. This week is the baseline and it is spoiled if you intervene.

WEEK 2 · PLACING YOURSELF ON THE CURVE

- 1 Add up week 1's moderate minutes and add the vigorous ones multiplied by 2.
- 2 Compare with the 150 minutes of the recommendation. That ratio is your position on the graph.
- 3 If you are below, your objective this month is to get closer, and every minute you add is worth a lot.

- 4 If you are already above, your objective is not to add minutes: it is for the 2 days of strength to appear if they are missing, which is the commonest gap in people who already do aerobic work.

WEEK 3 · ONE VARIABLE, ONE STEP

- 1 Choose ONE thing to raise: minutes, strength days, or weight in one exercise. One.
- 2 Raise it by one small step and hold it for the whole week.
- 3 Write down how you arrive at the end of the week: sleep, appetite for the next session, aches lasting more than two days.
- 4 If any of those three clearly got worse, the step was too big. Lower it and repeat. This is not a failure: it is the information you were looking for.

WEEK 4 · AUDITING YOUR LIFTING TECHNIQUE

- 1 For a week, with every heavy object you lift —inside and outside the gym— check the two factors that weigh most: is the load close to the body? are you twisting your trunk?
- 2 Write down the times the answer was bad. It is almost always the same situations: taking something out of the car boot, picking a child up off the floor, moving a box off a high shelf.
- 3 Fix the worst of those situations by changing the logistics, not the posture: bring the box closer before lifting it, turn your feet instead of your trunk, lower the shelf.

- 4 Compare the first week with this one. If the minutes went up and the bad situations went down, the month was worth more than any single workout.

WHAT TO EXPECT FROM THE NOTEBOOK

A month does not change a body, and promising that would be a lie. What does change is that at the end you will have three facts you do not have now: how much you really do, where you fall on the curve, and which two or three everyday situations you lift badly in. With that, the next decision makes itself.

WHAT YOU DO NOT ASK OF A GUIDELINE

The close, and the limits. The physical activity guidelines are a public health document, and there are questions they do not answer and do not claim to.

FIVE THINGS THIS BOOK CANNOT GIVE YOU

- A programme for your case. The guidelines give a population framework. A previous injury, a condition, a competition or a specific aesthetic goal require someone to examine you.
- A promise about body composition. Exercise changes health very robustly and weight quite modestly; most of the change in weight is decided in the kitchen and not in the gym.
- An answer about sports supplements. They are outside this book's sources. What can legally be claimed about them in Europe is recorded in a public register of health claims, and it is considerably less than the advertising suggests: it is worth consulting before buying anything.
- A verdict on which method is best. Long sets or short, high frequency or low, machines or free weights: within a reasonable range, everything works if it is done with progression and consistency. The dominant variable is not the method: it is how much of it you actually do.

- Permission to ignore pain. Diffuse muscular discomfort lasting one or two days is normal. Localised joint pain, pain that always appears in the same movement, or pain that worsens session by session, is not.



GLOSSARY

AEROBIC WORK	Sustained activity that raises heart rate and breathing for a prolonged period.
UNLOADING	Absence of mechanical load on the skeleton. The term comes from spaceflight and bed rest research.
DETRAINING	Progressive loss of the adaptations on stopping training.
MET	Metabolic equivalent. A multiple of resting energy expenditure. One MET is roughly one kilocalorie per kilo per hour.
PROGRESSION	Gradual increase in some training variable. Without it, adaptation stops.
THE TALK TEST	A method for estimating relative intensity: talking yes and singing no amounts to moderate.
OVERUSE	Injury from the accumulation of repeated load without sufficient recovery, not from a one-off accident.
VOLUME	Total amount of work: sets times repetitions times weight, or total minutes in the aerobic case.



And with that the book closes. It has a dose with its curve, a scale for measuring intensity, eight patterns for auditing any week, six variables for progressing when the weight will not go up, an equation for lifting, a battery for measuring yourself and a description of what happens to a body when none of that is there.

What it does not have is a closed programme, and that is on purpose: the programme depends on your timetable, your equipment and your history. Almost everyone's problem is not choosing the right programme, but sticking to any of the reasonable ones for long enough. With what is in these pages you can build your own, audit it every week and correct it with data instead of impressions. That is worth more than any template.

VILLUMINATIONS



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